Project Report on

"MY WALLET – A Full Stack Expense Tracking Application"

Submitted in fulfillment of the requirements of
Post Graduate Diploma in Advanced Computing

SUBMITTED BY

AUGUST 2025

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CERTIFICATE

This is to certify that the Report work entitled

" My Wallet - Expense Tracking App"

Has been duly completed by the following students under my guidance, in a satisfactory manner as a fulfillment of the requirements for the award of the PG- Diploma in Advanced Computing, Delhi

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Declaration

I declare that this written submission represents our ideas in our own words and where others' ideasor words have been included, we have adequately cited and referenced the original sources. We also declarethat we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of theabove will be cause for disciplinary action by the Institute and can also evoke penal action from the sourceswhich have thus not been properly cited or from whom proper permission has not been taken when needed.

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Date : 23/02/2026

ACKNOWLEDGEMENT

We are heartily thankful to our guide Prof. **Pankaj Kumar Sir**, for his guidance, patience, and support . We consider ourselves very fortunate for being able to work with very considerate and encouraging professor like him. Without his guidance to complete these study work, we would not finished our project. It is also our duty to record gratitude to our whole CDAC department for their help in needs.

Our special thanks to our parents and all of friends for helping us exchange ideas and give the enjoyable study environment . At last we special gratify to almighty God for blessing us with the hidden power to completing this study work.

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ABSTRACT

The rapid growth of digital transactions has increased the need for effective personal finance management systems. “My Wallet – A Full Stack Expense Tracking Application” is designed to provide a secure, reliable, and user-friendly platform for managing income, expenses, budgets, and financial analytics.

The application allows users to record and categorize transactions, monitor budgets, manage recurring payments, and visualize spending patterns through interactive dashboards. It also includes an administrator module that enables centralized management of users, categories, and transactions.

The system is developed using modern technologies such as React.js for the frontend, Spring Boot for the backend, and MySQL for database management. Security is ensured through JWT-based authentication, encrypted passwords, and role-based access control.

This project aims to simplify financial tracking, improve financial awareness, and provide a scalable solution for personal expense management. The system is efficient, secure, and adaptable for future enhancements such as mobile applications and advanced analytics.

Table of Contents

PROJECT SYNOPSIS

Title of the Project

MY WALLET – A Full Stack Expense Tracking Application

Introduction

"My Wallet" is a comprehensive full-stack expense tracking application designed to help individuals manage their personal finances efficiently and securely. The application enables users to record, categorize, and analyze their income and expenses while providing administrators with complete oversight of the system. Built using modern web technologies, the system ensures data accuracy, security, and scalability.

Problem Statement

Managing personal finances manually or using traditional tools such as notebooks and spreadsheets is inefficient, error-prone, and lacks security. Many existing applications provide limited features and do not support advanced analytics, role-based access, or centralized monitoring. There is a need for a secure, user-friendly, and feature-rich financial management system.

Proposed Solution

"My Wallet" provides a centralized web-based solution that allows users to track daily transactions, set budgets, manage recurring payments, and visualize financial data through interactive dashboards. An admin module ensures effective management of users, categories, and transactions, thereby maintaining system integrity.

Objectives of the Project:

- To develop a secure and reliable expense tracking system
- To help users manage income, expenses, and budgets efficiently
- To provide visual insights into financial data
- To implement role-based authentication and authorization
- To enable administrators to monitor and manage system activities

Technologies Used:

- **Backend:** Spring Boot, Spring Security, JWT
- **Frontend:** React.js, CSS3
- **Database:** MySQL
- **Tools:** Maven, Node.js, Axios

Scope of the Project:

The application is suitable for individuals and small organizations for managing financial records. The system is scalable and can be enhanced with mobile applications, advanced analytics, and third-party integrations in the future.

CHAPTER - 1

INTRODUCTION

Effective financial management is essential for maintaining financial stability. Improper expense tracking often leads to unnecessary spending and poor financial planning. "My Wallet" addresses these issues by offering a digital platform that simplifies financial record management.

The application is developed using a client-server architecture, where the frontend is implemented in React.js and the backend is built using Spring Boot. Secure authentication and authorization are implemented using JWT, ensuring data confidentiality and system reliability.

CHAPTER- 2

EXISTING SYSTEM

Existing System:

The existing methods of expense management include manual bookkeeping, spreadsheets, and basic expense tracking applications. These systems provide limited functionality and lack advanced features such as budget tracking, analytics, and centralized administration.

Limitations of the Existing System

- High probability of human errors
- Lack of data security
- No role-based access control
- Limited financial analysis capabilities
- Absence of centralized monitoring

Need for the Proposed System

A modern, secure, and scalable expense tracking system is required to overcome the limitations of traditional methods and provide meaningful financial insights.

CHAPTER – 3

PROPOSED SYSTEM

System Overview

"My Wallet" is a full-stack web application that supports multi-role access for users and administrators. The system enables efficient financial tracking and management through a structured and secure platform.

Features of the Proposed System

User Module

- Secure registration and login
- Income and expense management
- Budget creation and tracking
- Recurring transaction management
- Financial statistics and visual reports

Admin Module

- User management
- Category management
- Transaction monitoring
- Search, filter, and pagination functionalities

Advantages of the Proposed System

- Enhanced security through JWT authentication and password encryption
- User-friendly and responsive interface
- Centralized data storage
- Real-time financial insights

CHAPTER – 4

SYSTEM DESIGN AND IMPLEMENTATION

4.1 System Architecture

The application follows a client-server architecture consisting of:

- **Frontend:** React.js
- **Backend:** Spring Boot REST APIs
- **Database:** MySQL

Backend Implementation

The backend is developed using Spring Boot and Spring Security. JWT is used for authentication and authorization. JPA repositories handle database operations, while Spring Mail is used for email verification and password recovery.

Frontend Implementation

The frontend is developed using React.js with reusable components. Axios is used for communication with backend APIs, and Context API is used for state management.

Database Design

The database consists of structured tables such as:

- Users
- Transactions
- Categories
- Budgets

These tables ensure efficient storage and retrieval of financial data.

4.2 Entity Relationship (ER) Diagram

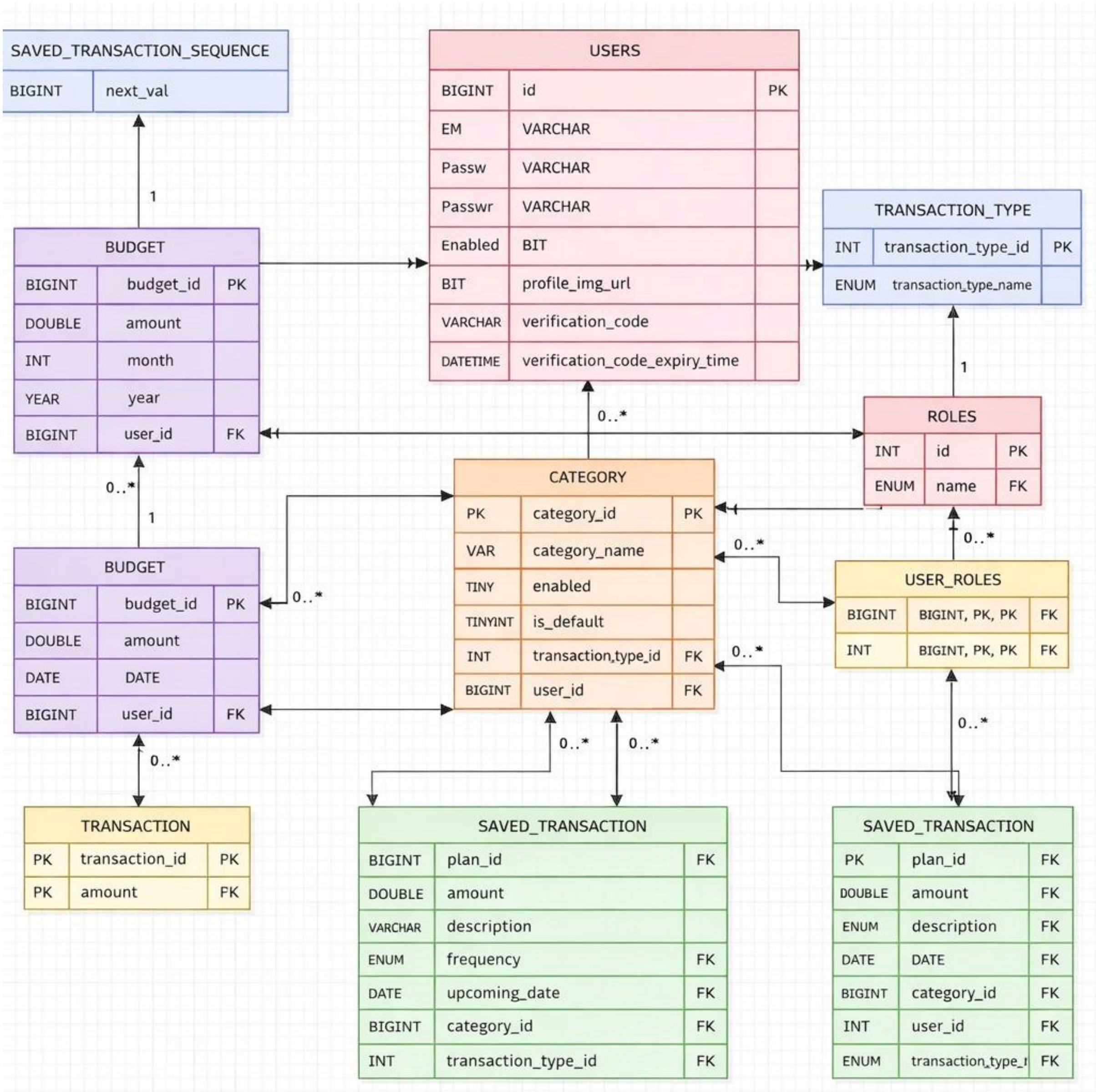

Figure 4.2: Entity Relationship Diagram

ER Diagram Description

The Entity Relationship (ER) Diagram represents the logical database structure of the **"My Wallet – Full Stack Expense Tracking Application."** It illustrates the entities involved in the system, their attributes, and the relationships among them. The diagram is designed to ensure data consistency, integrity, and efficient management of financial records.

The primary entity in the system is the **Users** table, which stores user-related information such as user ID, email, username, password, profile image URL, account status, and verification details. Each user can perform multiple financial operations within the system.

The **Transaction** entity stores individual income and expense records, including transaction amount, date, description, and references to the associated user and category. A **one-to-many relationship** exists between Users and Transactions, where a single user can have multiple transactions, but each transaction belongs to only one user.

The **Category** entity defines various expense and income categories such as food, salary, education, and healthcare. Each transaction is linked to a specific category, establishing a **many-to-one relationship** between Transactions and Categories. Categories can be user-defined or system-defined and may be enabled or disabled as required.

The **Budget** entity is used to store budget limits assigned by users for specific months and years. It includes attributes such as budget amount, month, year, and user reference. A user can define multiple budgets over time, resulting in a **one-to-many relationship** between Users and Budgets. Each budget may be associated with one or more categories.

The system also supports recurring and scheduled transactions through the **Saved Transaction** entity. This entity stores planned transactions with attributes such as frequency, upcoming date, transaction type, amount, and category. These transactions are linked to users and categories, enabling automated reminders and recurring expense tracking.

Role-based access control is implemented using the **Roles** and **User_Role** entities. The Roles entity defines system roles such as USER and ADMIN, while the User_Role table establishes a **many-to-many relationship** between users and roles. This design ensures secure authorization and controlled access to system features.

Overall, the ER diagram demonstrates a normalized and scalable database design that supports secure user management, detailed transaction tracking, budgeting, recurring payments, and administrative control. The structured relationships among entities enable efficient data storage, retrieval, and future system expansion.

CHAPTER – 5

RESULTS AND DISCUSSION

The system was successfully implemented and tested. Users can efficiently manage their income and expenses, monitor budgets, and analyze financial trends through visual dashboards. The admin module enables effective system monitoring and control.

The application demonstrates high performance, secure data handling, and ease of use, fulfilling all defined project objectives.

CHAPTER – 6

CONCLUSION AND FUTURE SCOPE

Conclusion:

“My Wallet” is a secure, efficient, and user-friendly expense tracking application that simplifies personal financial management. The system effectively integrates modern technologies to provide reliable and scalable financial solutions.

Future Enhancements:

- Development of a mobile application using React Native
- Export of financial data in PDF, Excel, and CSV formats
- AI-based financial analysis and recommendations
- Integration with banking APIs
- Multi-language support and dark mode

CHAPTER - 7

SYSTEM IMPLEMENTATION AND OUTPUT SCREENS

7.1 Introduction

This chapter presents the implementation details and output screens of the **“My Wallet – Full Stack Expense Tracking Application.”** It demonstrates the successful execution of the system and verifies that all functional requirements have been implemented as intended.
The screenshots included in this chapter represent real-time system outputs generated during application usage. They illustrate how different modules such as authentication, transaction management, budgeting, analytics, and administrative controls function in an integrated environment. This chapter serves as practical evidence of system usability, reliability, and performance.

7.2 Authentication Module

The authentication module ensures secure access to the system through user registration and login mechanisms. It implements email verification, encrypted passwords, and role-based access control to protect sensitive financial data.

7.2.0 Welcome Page

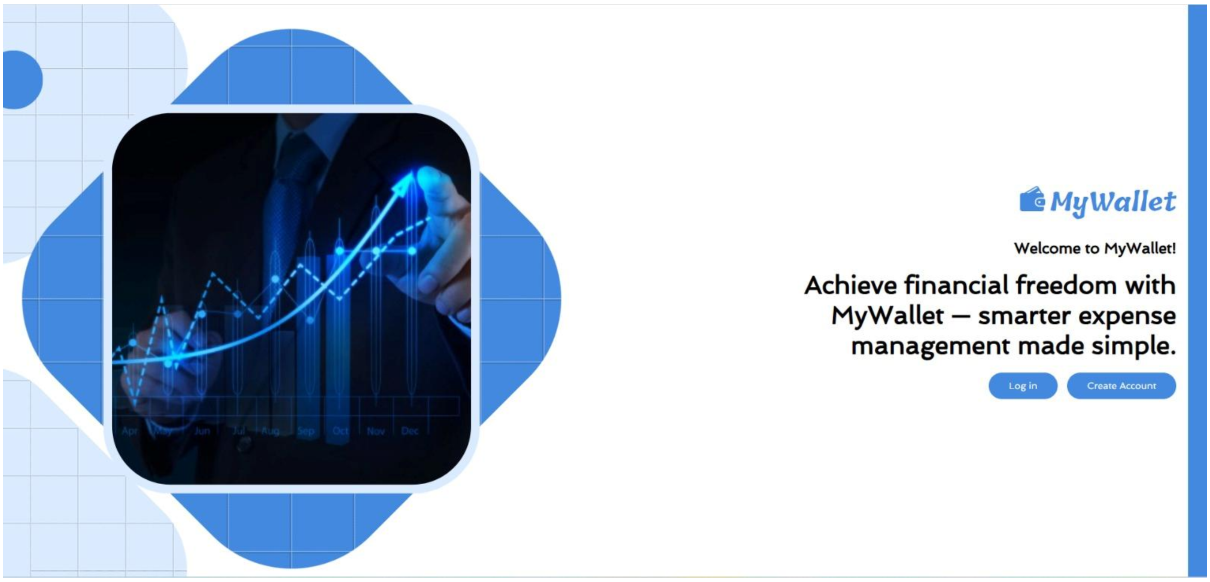

Figure 7.0: Welcome Page of My Wallet Application

The Welcome Page serves as the initial interface of the “My Wallet” application. It provides a brief introduction to the system and highlights its core purpose of effective financial management. The page includes call-to-action options such as Login and Create Account, allowing users to proceed with authentication. The clean and user-friendly design ensures a positive first impression and smooth navigation.

7.2.1 Login Page

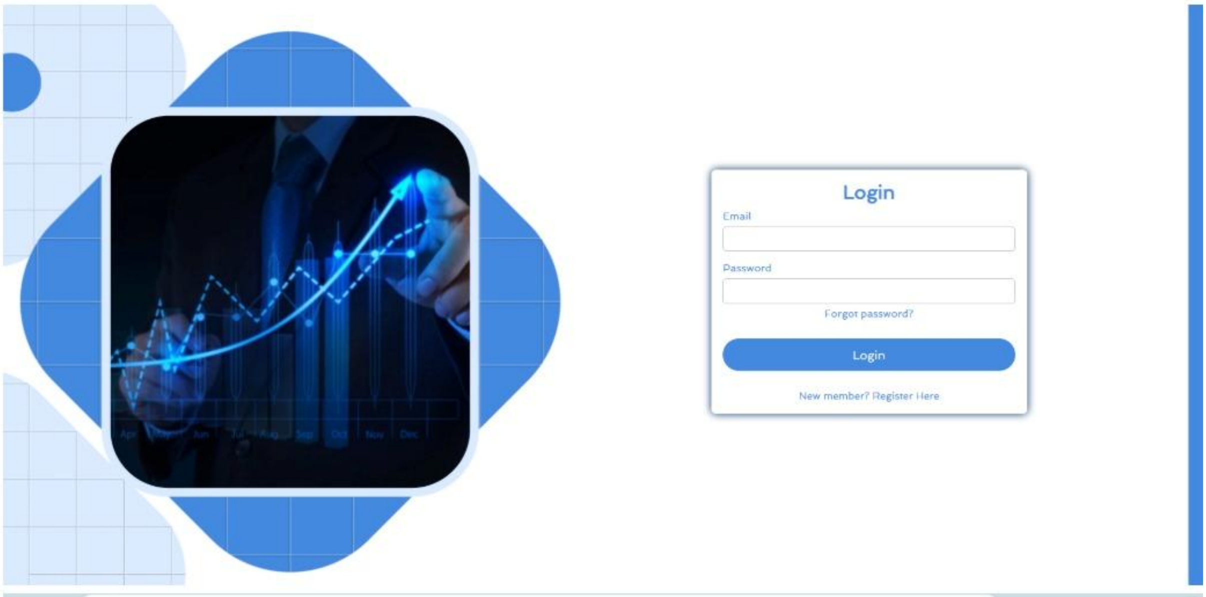

Figure 7.1: User Login Page

The login page allows registered users to access the system by entering valid credentials. The authentication process is secured using JWT-based authorization, ensuring that only authenticated users can access protected resources. Error handling mechanisms prevent unauthorized access and improve system reliability.

7.2.2 Registration Page

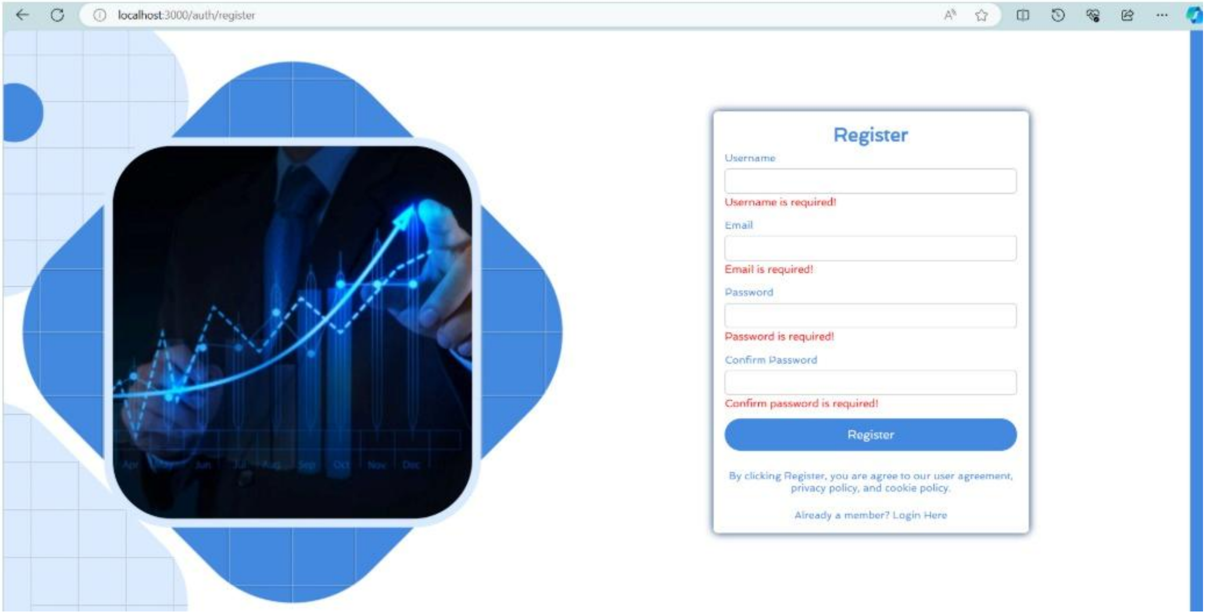

Figure 7.2: User Registration Page

This page enables new users to create an account by submitting required information such as email address, username, and password. Email verification is mandatory to activate the account, which enhances security and prevents unauthorized registrations.

7.3 User Dashboard and Transaction Management

This module provides users with complete control over their financial records. It includes dashboard visualization, transaction history, and transaction creation functionalities.

7.3.1 Dashboard Overview

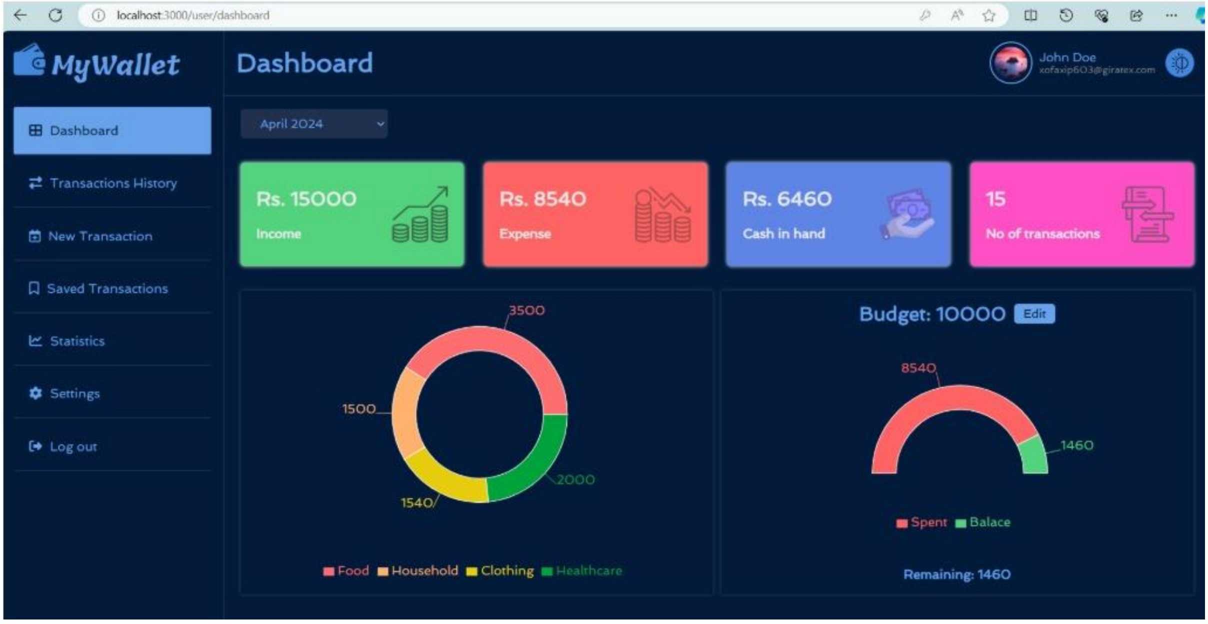

Figure 7.3: User Dashboard

The dashboard presents a consolidated overview of the user's financial status, including total income, total expenses, available cash balance, and number of transactions. Graphical representations assist users in understanding their spending behavior efficiently.

7.3.2 Transaction History

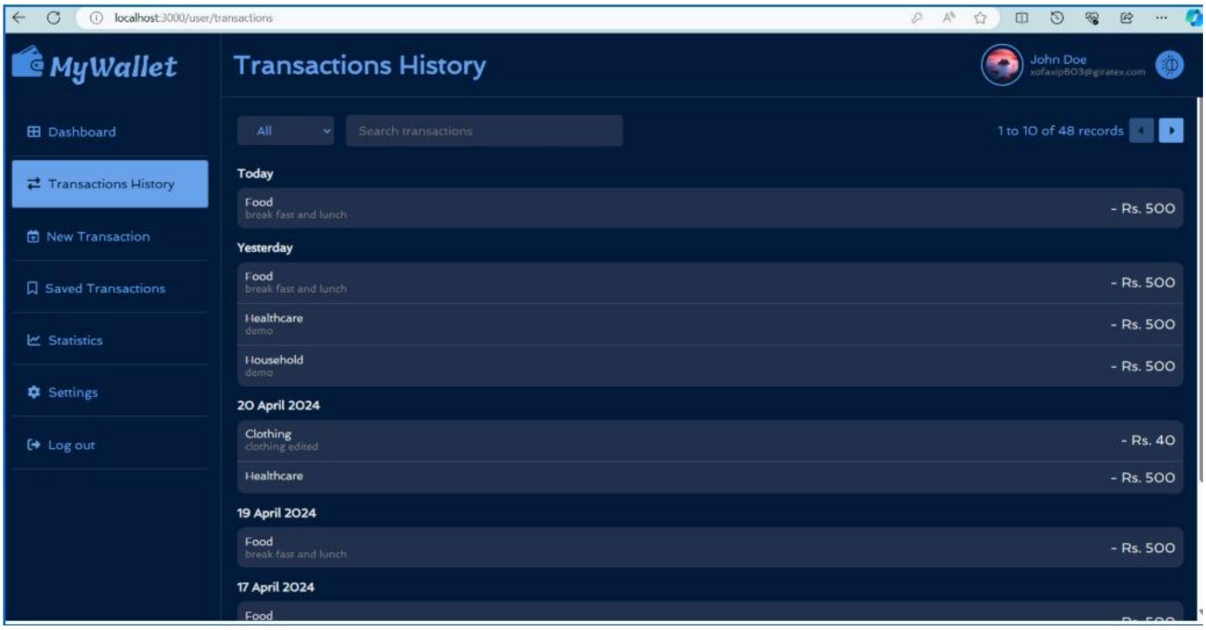

Figure 7.4: Transaction History Page

This screen displays all recorded transactions in a structured format. Users can search, filter, and review transactions based on date, category, and transaction type. This feature ensures transparency and accurate financial tracking.

7.3.3 Add New Transaction

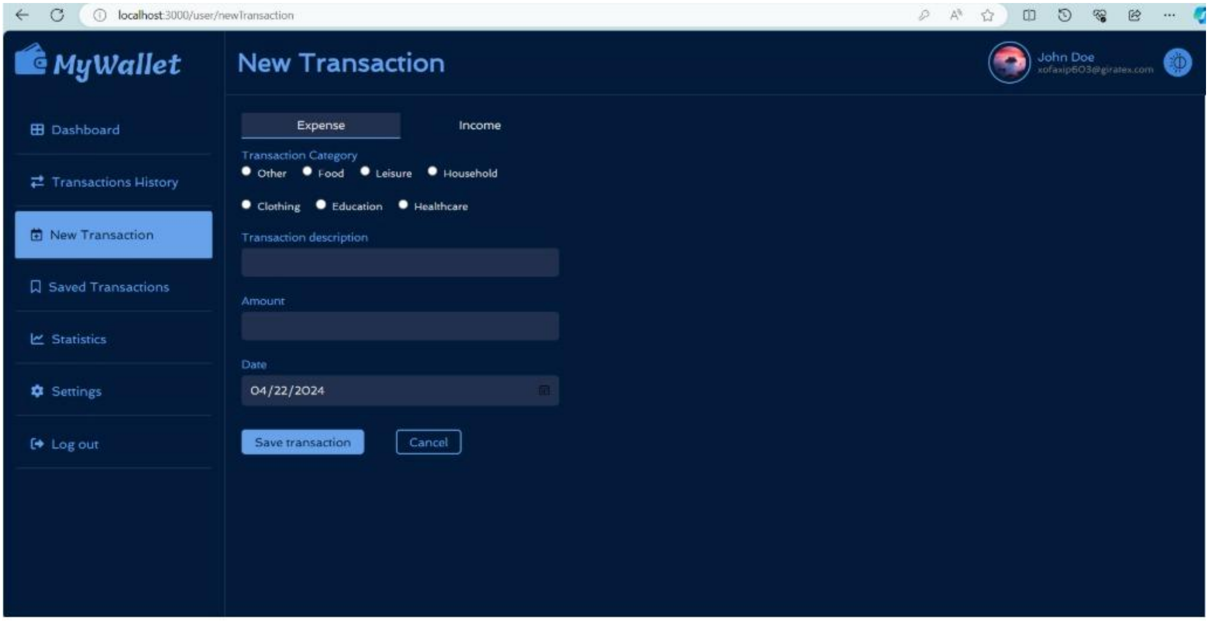

Figure 7.5: Add New Transaction Page

The add transaction screen allows users to record new income or expense entries by selecting appropriate categories and entering transaction details. Input validation ensures accuracy and data consistency before storing records in the database.

7.4 Recurring Transactions and Analytics

This module enhances financial planning by supporting recurring transactions and providing analytical insights.

7.4.1 Saved Transactions

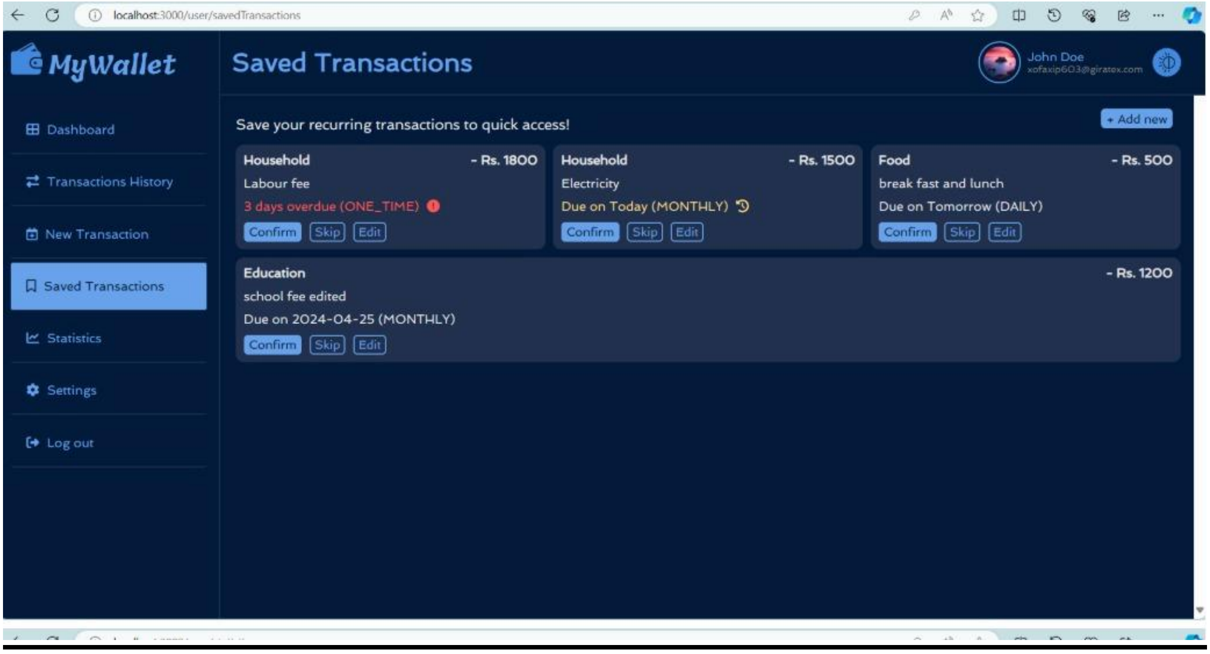

Figure 7.6: Saved Transactions Page

This screen manages recurring and scheduled transactions such as monthly bills and subscriptions. Users receive reminders for upcoming payments, which helps in maintaining consistent financial discipline.

7.4.2 Statistics and Analytics

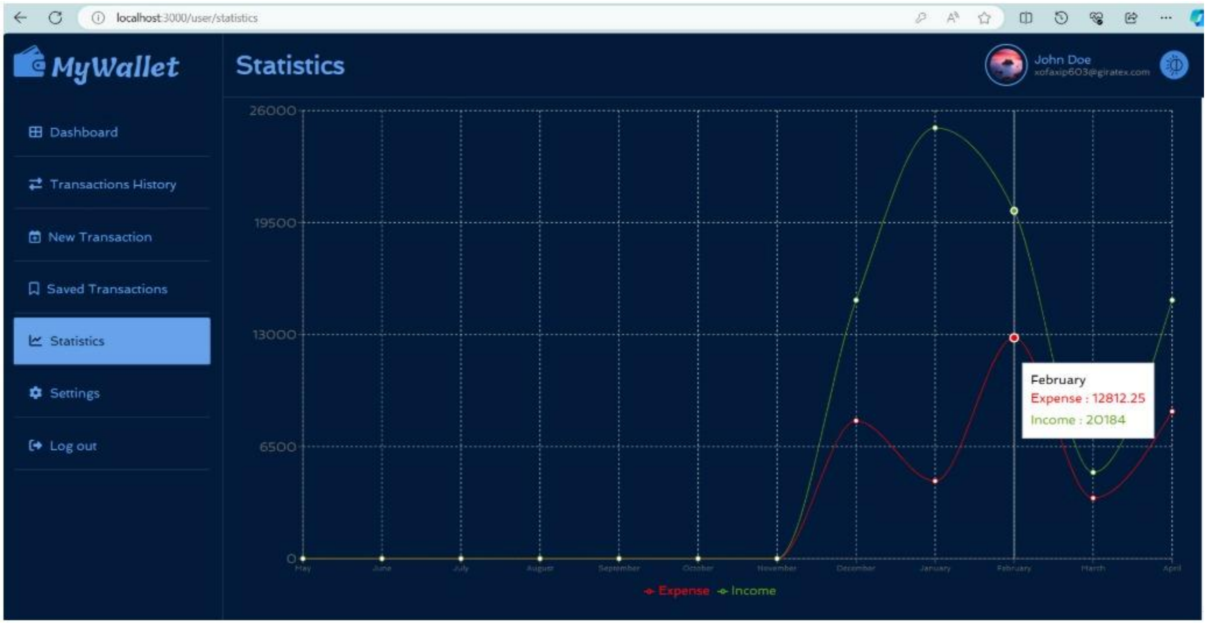

Figure 7.7: Statistics and Analytics Page

This page presents visual analytics in the form of charts and graphs that display income and expense trends over time. These insights support informed financial decision-making and long-term planning.

7.5 User and Administrative Management

This module highlights both user-level configuration and administrative control features.

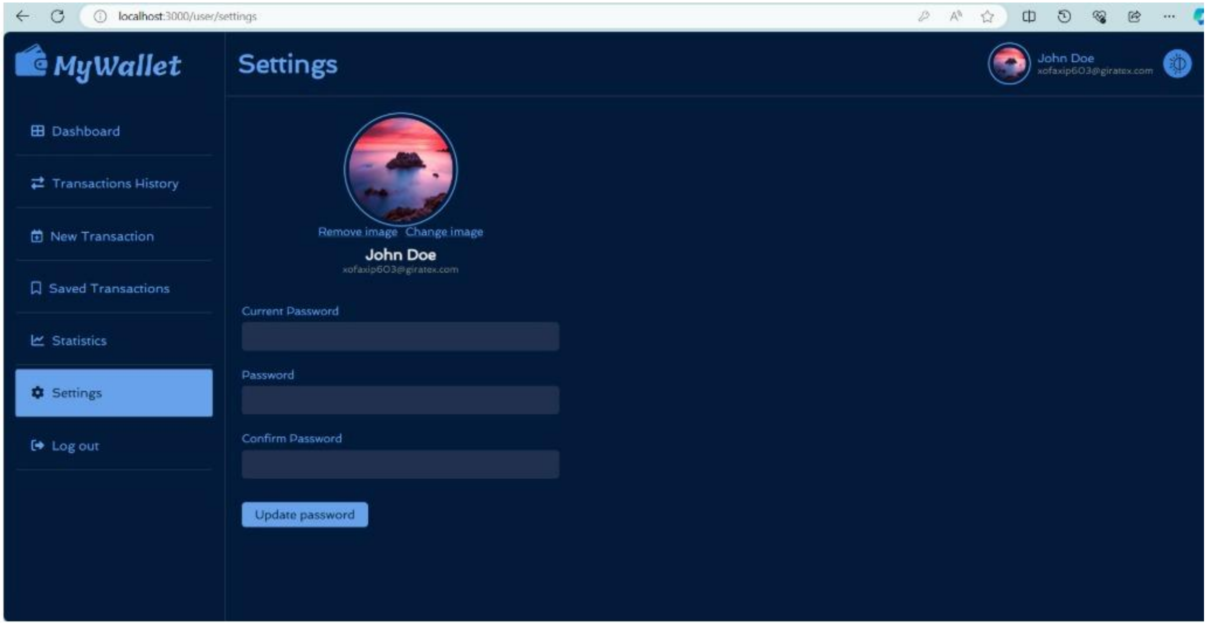

Figure 7.8: User Settings Page

The settings page allows users to update profile information, manage passwords, and customize account preferences. These features enhance personalization and account security.

7.5.2 User Management

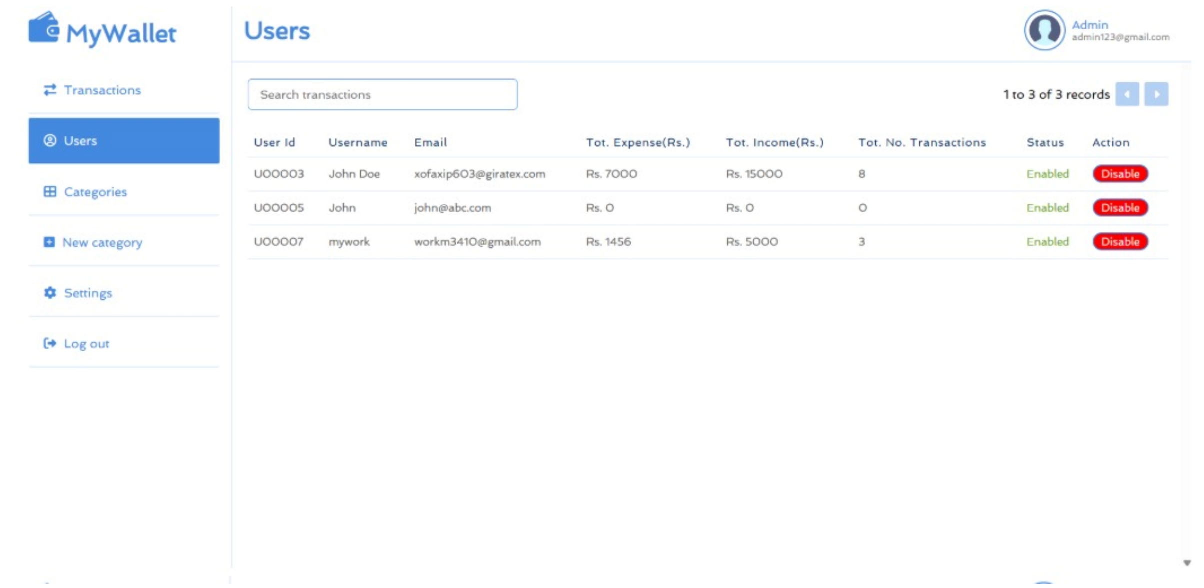

Figure 7.9: Admin User Management Page

This administrative screen enables administrators to manage registered users effectively. Admins can view user activity, enable or disable accounts, and ensure policy compliance.

7.5.3 Category Management

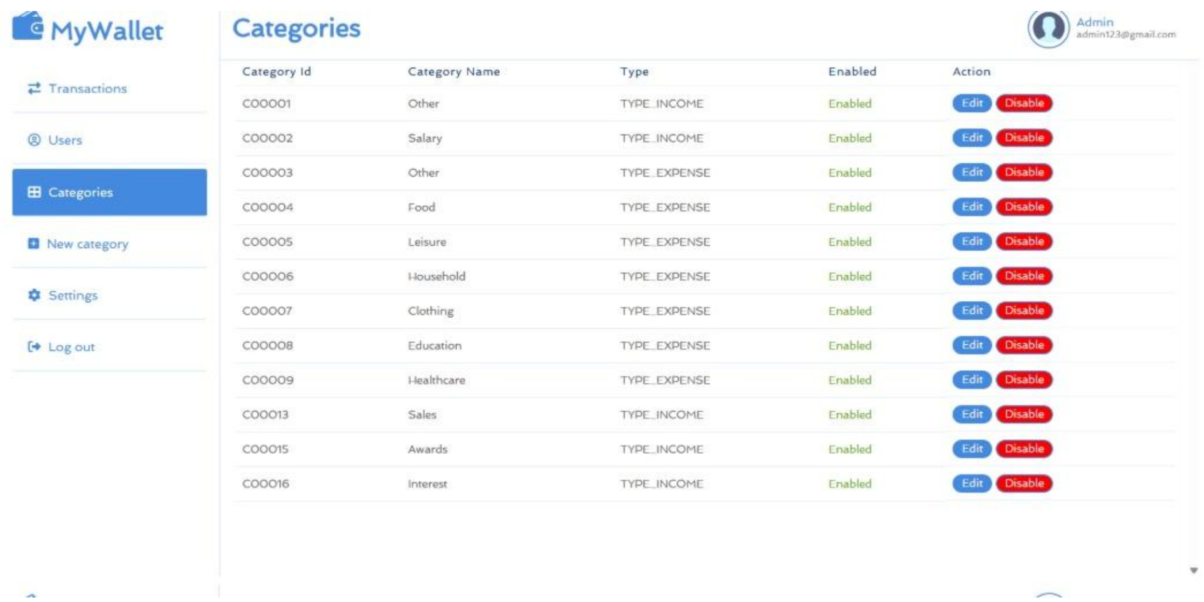

Figure 7.10: Admin Category Management Page

This page allows administrators to manage expense and income categories used throughout the system. Proper category management ensures data consistency and structured financial reporting.

7.6 Summary

This chapter demonstrated the successful implementation of the “My Wallet” application through various output screens. The screenshots confirm that all functional modules operate as intended and provide a seamless user experience. The system effectively integrates security, usability, and performance, validating the overall project implementation.

CHAPTER - 8

TESTING AND VALIDATION

8.1 Introduction

Testing is a critical phase in the software development lifecycle that ensures the reliability, correctness, and performance of the system. This chapter describes the testing strategies and validation techniques applied to the "My Wallet – Full Stack Expense Tracking Application." The objective of testing is to verify that the system meets all functional and non-functional requirements and operates as expected under different scenarios.

8.2 Testing Strategy

The application was tested using a combination of manual testing and functional testing techniques. Each module was tested independently and then validated as part of the integrated system. Testing was performed at both frontend and backend levels to ensure smooth interaction between components.
The following testing approaches were used:

- Unit Testing
- Integration Testing
- System Testing
- Security Testing

8.3 Unit Testing

Unit testing focuses on verifying individual components of the system. Backend services such as user authentication, transaction processing, category management, and budget calculations were tested using sample inputs to ensure correct outputs.
Frontend components such as forms, dashboards, and navigation elements were also tested to validate user interactions and error handling.

8.4 Integration Testing

Integration testing was performed to ensure seamless communication between the frontend and backend modules. API endpoints were tested using real-time data to validate request handling, response accuracy, and error management. This phase confirmed that modules such as authentication, transaction management, and analytics work together without conflicts.

8.5 System Testing

System testing evaluated the complete application in a real-world environment. Scenarios such as user registration, login, transaction creation, budget tracking, recurring transactions, and admin management were tested end-to-end.
The system successfully handled concurrent users and maintained data consistency throughout operations.

8.6 Security Testing

Security testing was conducted to protect sensitive financial data. Password encryption using BCrypt, JWT-based authentication, role-based access control, and email verification mechanisms were validated to ensure system security. Unauthorized access attempts were prevented, and secure session management was confirmed.

8.7 Validation Results

The testing process confirmed that the application meets all specified requirements. All critical functionalities performed as expected, and no major defects were identified. Minor issues encountered during testing were resolved during development iterations.

8.8 Summary

This chapter demonstrated that the “My Wallet” application was thoroughly tested and validated. The testing process ensured system stability, security, and usability, making the application suitable for real-world financial management scenarios.

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